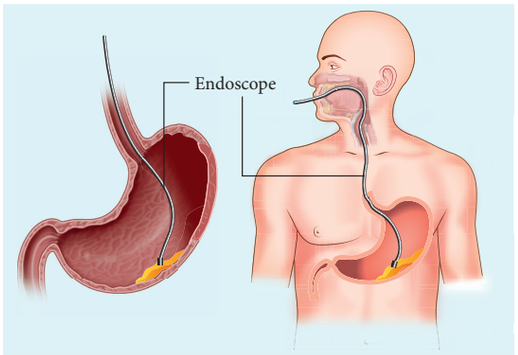
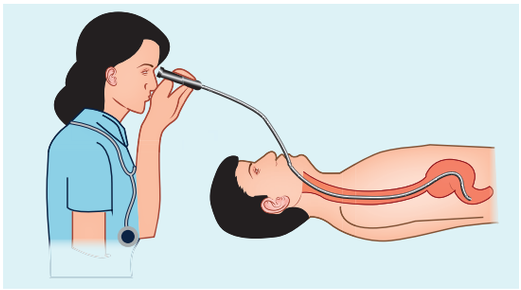
DO YOU KNOW?

An endoscope which has a bundle of optical fibres is an instrument used by doctors to see inside of a patient's body. Endoscopes work on the phenomenon of total internal reflection. The optical fibres are inserted into the body through mouth, nose (or) a special hole made in the body. Even operations could be carried out with the endoscope cables which have the necessary instruments attached at their ends.





6.4.8 Refraction in glass slab

When a ray of light enters a slab it travels from rarer medium (air) to denser medium (glass). This results in deviation of the ray towards the normal. When the light ray leaves the slab it travels from denser medium (glass) to rarer medium (air) resulting in deviation of the ray away from the normal. After the two refractions, the light ray emerges in the same direction as that of the incident ray on the glass slab with a lateral displacement (or) shift *L*. i.e. There is no change in the direction of the ray but, the path of the incident ray and refracted ray are parallel to each other with a shift L, as shown in Figure 6.30.

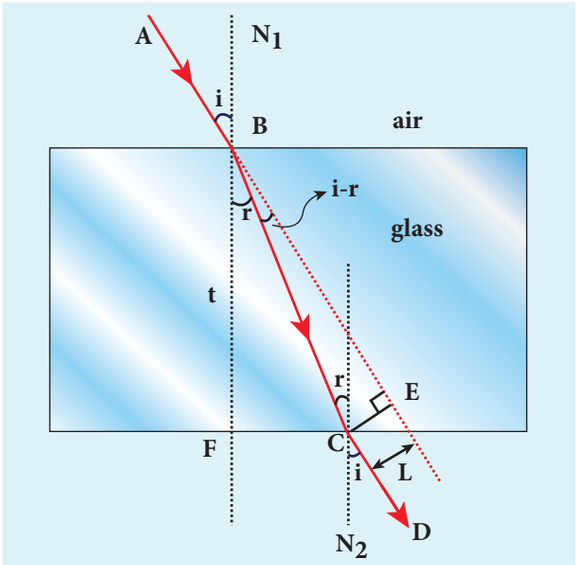


Figure 6.30 Refraction in glass slab

Consider a glass slab of thickness *t* and refractive index *n* kept in air medium. If the path of the light is *ABCD*, the refractions occur at two points *B* and *C* in the glass slab. The angles of incidence *i* and refraction *r* are measured with respect to the normal N_1 and N_2 at the two points *B* and *C* respectively. The lateral displacement *L* is the perpendicular distance *CE* drawn between the paths of the deviated light and the undeviated light at point *C*.

In the right angle triangle ΔBCE,

$$\sin(i-r)=\frac{L}{BC};\quad BC=\frac{L}{\sin(i-r)} \qquad (6.48)$$

In the right angle triangle ΔBCF,

$$\cos(r)=\frac{t}{BC};\quad BC=\frac{t}{\cos(r)} \qquad (6.49)$$

XII_Physics_Vol-2_UNIT-6_EM.indd 26 26-12-2022 11:38:07